

TripCentral

Social Travel Planning & Location Discovery App

CSE327.6 - Software Engineering Planning Document
Phase 1 — Requirements, Architecture & Design Specification

Technology Stack: Flutter • Supabase (PostgreSQL) • Google Maps Platform
Target Platform: Android (Open Source)
Design: Material 3 (Material You)

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Members

Name	ID
Faiyad Tanjeem Hoque	2322415042
Rafee Maisoon	2323012042
Md. Shahriar Rakib Rabbi	2321937642
Tazreen Alam	2321506042

Github Link: <https://github.com/Rafee-M/TripCentral>

1. Project Overview

1.1 Background:

Trip Central is a travel platform designed to turn the tedious chore of planning a trip into a real-time, collaborative social process. The goal of the app is to provide useful and relevant information to travellers, where the locations can be saved into a custom list, routes can be plotted for future use, and the changes can be synchronized across all the participants using an observer-based notification system.

Unlike static travel blogs or platforms like Yelp, TripCentral is built around the concept of trusted discovery. You plan with your friends, share your experience with the public, and discover places curated by real people in destinations you are exploring. The application aims to bridge the well-known gap in travel planning. It will remove hassles of a scattered planning process and help users decide where to go, organise the journey in a structured collaborative list format with blocks, categories, and attached reviews. Everyone added to a list can evaluate locations with detailed information, collaborate in a structured manner, and get a visualisation of the trip.

1.2 Tech Stack:

From a technical standpoint, TripCentral will be built around the Flutter UI framework, ensuring a single codebase that compiles natively to Android. It also makes future scalability easier as the Dart programming language compiles natively to multiple platforms, including iOS, web, Windows, macOS, and Linux. This drastically reduces development time, effort, and maintenance costs.

Supabase will provide the complete backend infrastructure, authentication, a PostgreSQL relational database with Row Level Security (RLS). It also provides built-in authentication (email/password, OAuth), realtime database subscriptions via WebSockets, file/image object storage, and a free tier (50,000 monthly active users, 500MB database, 1GB storage). Critically, Supabase is itself open-source, which aligns perfectly with the project's ethos.

Google Maps Platform drives location search, enriched place data, ratings, photos, and map rendering. Relevant APIs include:

API	Purpose in TripCentral	Key Endpoint / Method	Free Tier (Monthly)
Places API (New) — Text Search	Search for locations by name/keyword in the list creation search bar	POST /v1/places:searchText	Essentials: generous free quota
Places API (New) — Place Details	Fetch full details (hours, phone, website, reviews summary) for a tapped place	GET /v1/places/{place_id}	Essentials / Pro based on fields
Place Photos API	Display a photo thumbnail for each search result and list item	GET /v1/places/{name}/photos/{name}/media	Included with Place Details call
Autocomplete (New)	Predictive text suggestions while the user types in the search bar	POST /v1/places:autocomplete	Sessions are FREE when linked to a Place Details call
Maps SDK for Android	Render interactive map view inside the "Map View" feature for a list	Flutter: google_maps_flutter package	Dynamic Maps: free monthly quota

1.3 Feature Overview & Utility:

We solve a common problem that everyone faces in their lives, planning an outing/trip as a group. There's confusion on where to go, taking everyone's ideas into consideration and evaluating them. That's where TripCentral comes in with features such as:

- Lists: Acts like a collection of places that allows users to note down and store places they'd like to visit, change their sequences easily and compartmentalize ideas with sub-blocks.
- Google Maps Integration: The application will use Google Maps to interact with the global database of locations and extract all the relevant information in the list. Users can select individual points of interest and

insert them into the centralised list. They can easily click a place in their list and be greeted with an information card fetched from the Google Maps Places API, providing location details, pictures, reviews, schedules, etc., enhancing accessibility.

- Sharing and Collaboration: Users can share their list with friends and allow them to collaborate by proposing or directly modifying places, sequence, etc. Each list will have its own integrated group chat, allowing users to discuss plans in detail, directly share locations within the chat and keep these discussions organized. Therefore, decisions made in chat can be edited immediately in the itinerary list. All modifications to the list are notified to collaborators immediately so no one is left out of the loop.
- Discovery: Users can choose to make their lists public to anyone. Through a centralized search system with tag filters, users can search for existing trip plans at specific locations, restaurant recommendations for a specific cuisine, etc. driven completely by community spirit to share knowledge and joy.

1.4 End Users & Actors:

Primary Actors

1. **Authenticated User (Core Actor)**: A registered, email-verified user. This is the primary persona the app is designed for. They can create lists, add locations, connect with friends, receive notifications, share lists publicly, and submit reviews.
2. **Guest / Unauthenticated Visitor**: A user who has not logged in. They can browse publicly shared lists and view location details, but cannot create lists, leave reviews, or connect with other users. This actor is important for discoverability and acquisition.
3. **Connected User (Secondary Actor)**: An authenticated user who has an approved connection with another user. They can view each other's non-public lists if shared, receive notifications about trip edits, and be invited to collaborative lists.

4. **System (Automated Actor):** The backend system itself. It triggers email verification on sign-up, dispatches in-app notifications when friend requests are sent or trips are edited, scores and ranks public lists, and aggregates user reviews with Google Maps data.

1.5 Functional Requirements:

The functional requirements below are expressed as capabilities the system MUST provide. They are numbered for traceability.

FR-01: Authentication & Account Management

1. Authentication - The user will be able to login and sign up while also having a mandatory verification step to ensure account integrity.
2. Verification - The system shall send a verification email upon registration. The account will remain in a “pending verification” state until confirmed.
3. Recovery - The system shall allow the user to reset their password with appropriate verification

FR-02: Maps Integration

4. Google Maps api - Serving up location data, images, reviews, and other relevant info to aid in planning via Google Maps **Places API**
5. List - Saving points of interest in a custom-named list for trip planning. There will also be another category of lists meant to recommend places.

FR-03: List Management & Collaboration

6. Sharing - Users can share their planned trip privately with selected individuals.
7. Place Discovery - The system shall allow users to search for locations via an integrated search bar that calls the Google Maps Places Text Search API. The system shall display place name, address, rating, and a photo thumbnail from the Google Maps Places API in search results.
8. Notes - The system shall allow users to add a search result as an item in a list, with optional personal notes attached.

9. Modifications - The system shall allow users to reorder items within a list via drag-and-drop. The system shall allow users to delete items from a list. The system shall allow users to delete or archive a list.
10. Invitations - Users can invite other users to a TripList.
11. Chat system - A chat system will help the users to discuss the planning and logistics of the trip.
12. Permission control - List creators can control who can directly change the list and others can propose changes that have to be merged into the list via approval
13. List Discovery - Users can choose to make their lists public as a form of recommendation and a public resource. When someone else is making a list, they can search for public lists (with tags to filter lists based on relevance and ranking) and users will be able to incorporate and subsequently rank those lists for future considerations
14. Rating - Users can rate the other user's planned trip.

FR-04: Notification System

15. The system shall maintain a persistent in-app notification inbox for each user.
16. The system shall generate a notification when a user receives an invitation to a list.
17. The system shall generate a notification when a list shared with the user is edited

FR-06: Location Reviews

18. The system shall allow authenticated users to write a text review and submit a star rating for any location
19. The system shall display TripCentral reviews alongside Google Maps reviews on a location detail screen
20. The system shall prevent users to create more than one review per location
21. Users will be able to edit and delete their reviews

1.6 Non-Functional Requirements:

NFR-01: Performance

1. **Concurrency:** The system should allow multiple users to simultaneously edit on a single shared list without data loss or conflicts
2. **Latency:** Chat messages and map interactions should have a low response time as low as 200ms to provide users a real-time experience.

NFR-02: Scalability & Reliability

3. **Scalability:** Observer pattern must be used efficiently in the architecture to scale notifications to thousands of concurrent users without reducing server performance.
4. **Backend Infrastructure:** Supabase's free tier supports up to 50,000 monthly active users; the architecture shall be designed to migrate to a paid tier gracefully as user numbers grow.
5. **Availability:** The system should maintain a 99.9% uptime, since travelers from different time zones will be relying on the app while in transit.

NFR-03: Security

6. **Security:** All user data, especially login history, search history, private chat logs, private trip history, must be encrypted. Private lists must remain invisible to anyone not explicitly invited to view them.
7. **API Key Security:** API keys (Google Maps, Supabase) shall never be committed to version control. They shall be injected at build time.
8. All communication between the app and backend shall occur over HTTPS/TLS.

NFR-04: Usability & Accessibility

9. **Usability:** The user interface should be self-explanatory and intuitive enough for new users to navigate easily to create a list and invite a friend within 60 seconds.
10. **Design:** The app shall follow Material 3 design guidelines throughout, ensuring a familiar and intuitive Android experience.

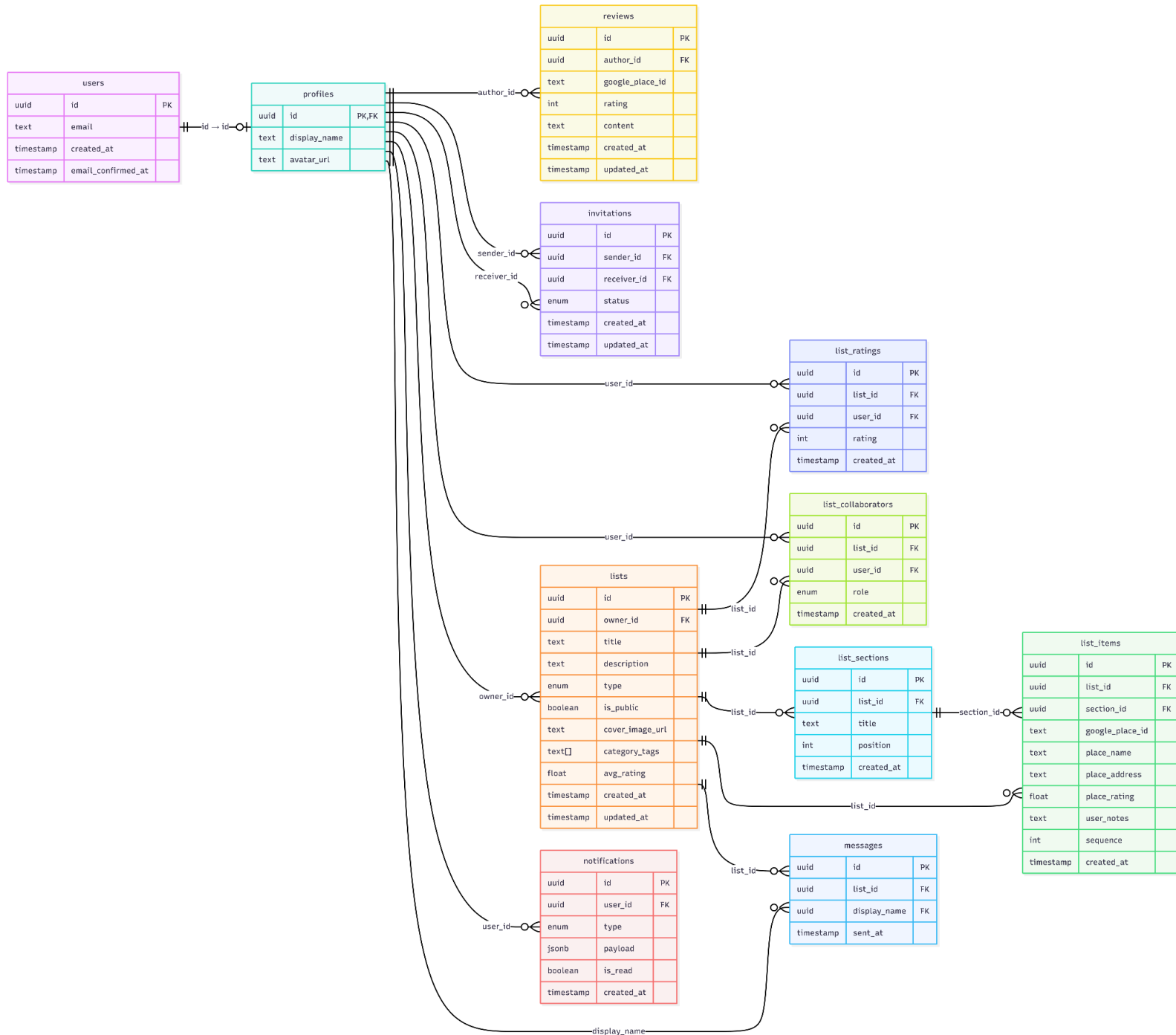
1.7 Conclusion:

Our project “Trip Central” represents a significant step towards transforming travel planning from a solitary, fragmented process into a cohesive, communal experience. By seamlessly blending real-time collaborative editing with social discovery tools, this platform eliminates the logistic friction that often hinders group travels. Ultimately, this project does more than just organise trips; it fosters connection by providing a shared space for friends to co-create experiences and for the wider community to discover high-quality routes through user’s evaluation. As travel continues to evolve into a more social and shared endeavour, the Trip Central stands ready to serve as the definitive digital companion for those who believe that the best journeys are the ones planned and experienced together.

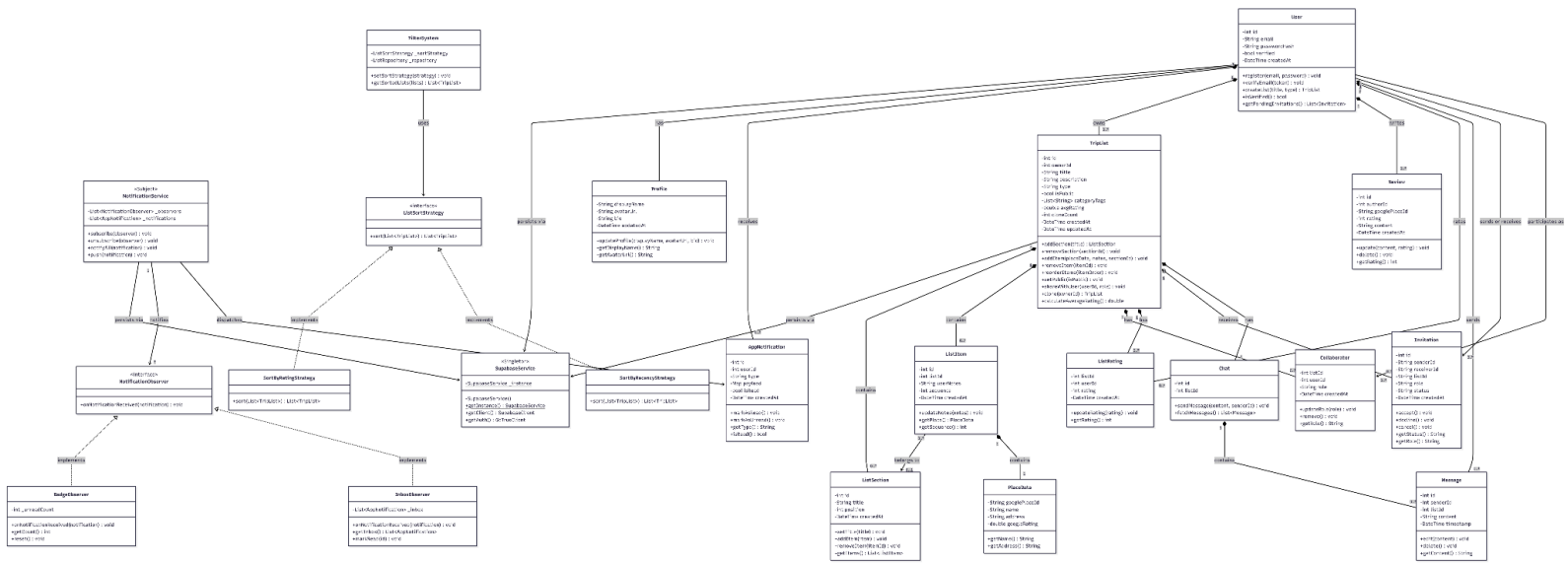
2. HLD Diagrams

All high resolution images are available [here \(Google Drive\)](#)

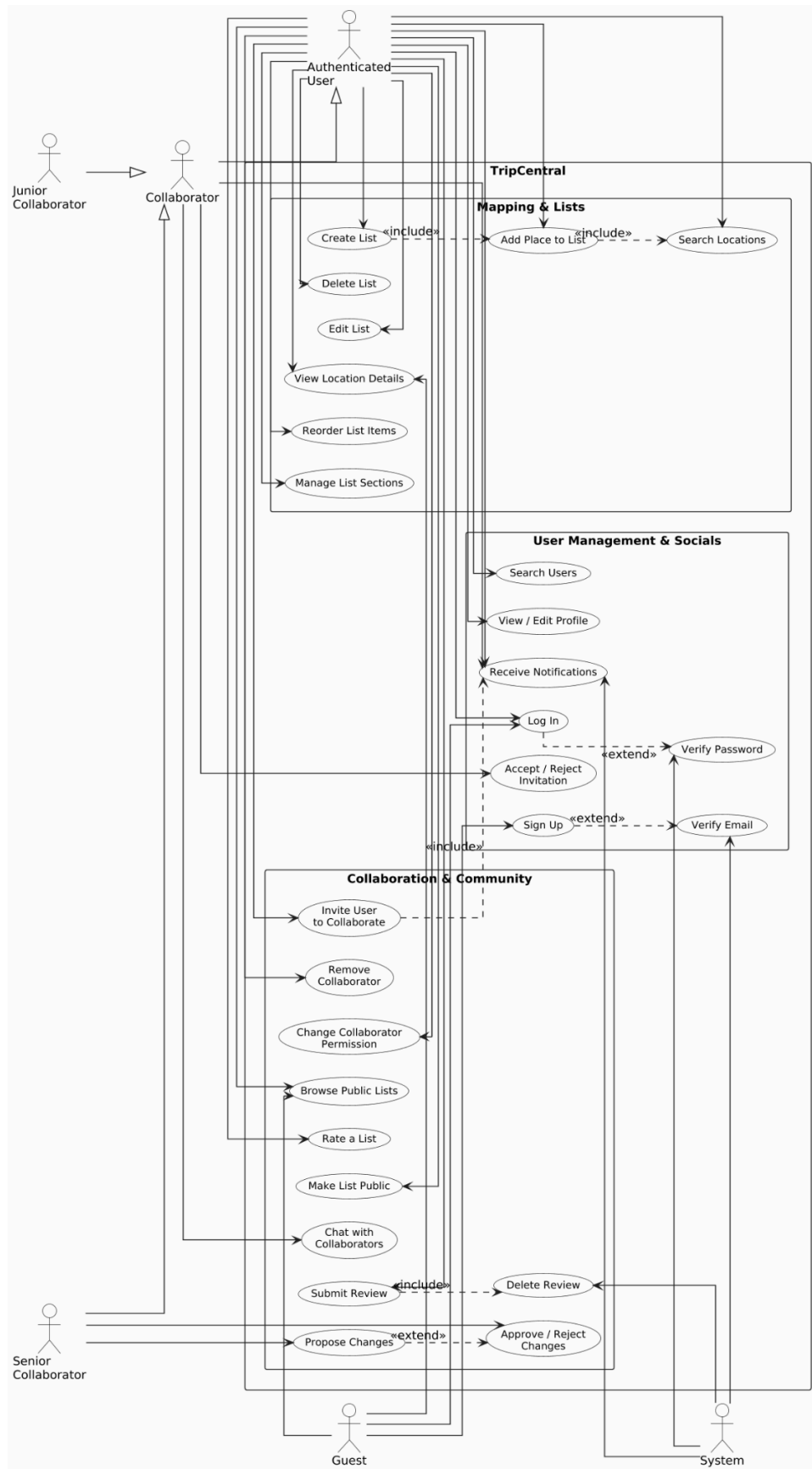
2.1 E-R Diagram:



[HD Image](#)



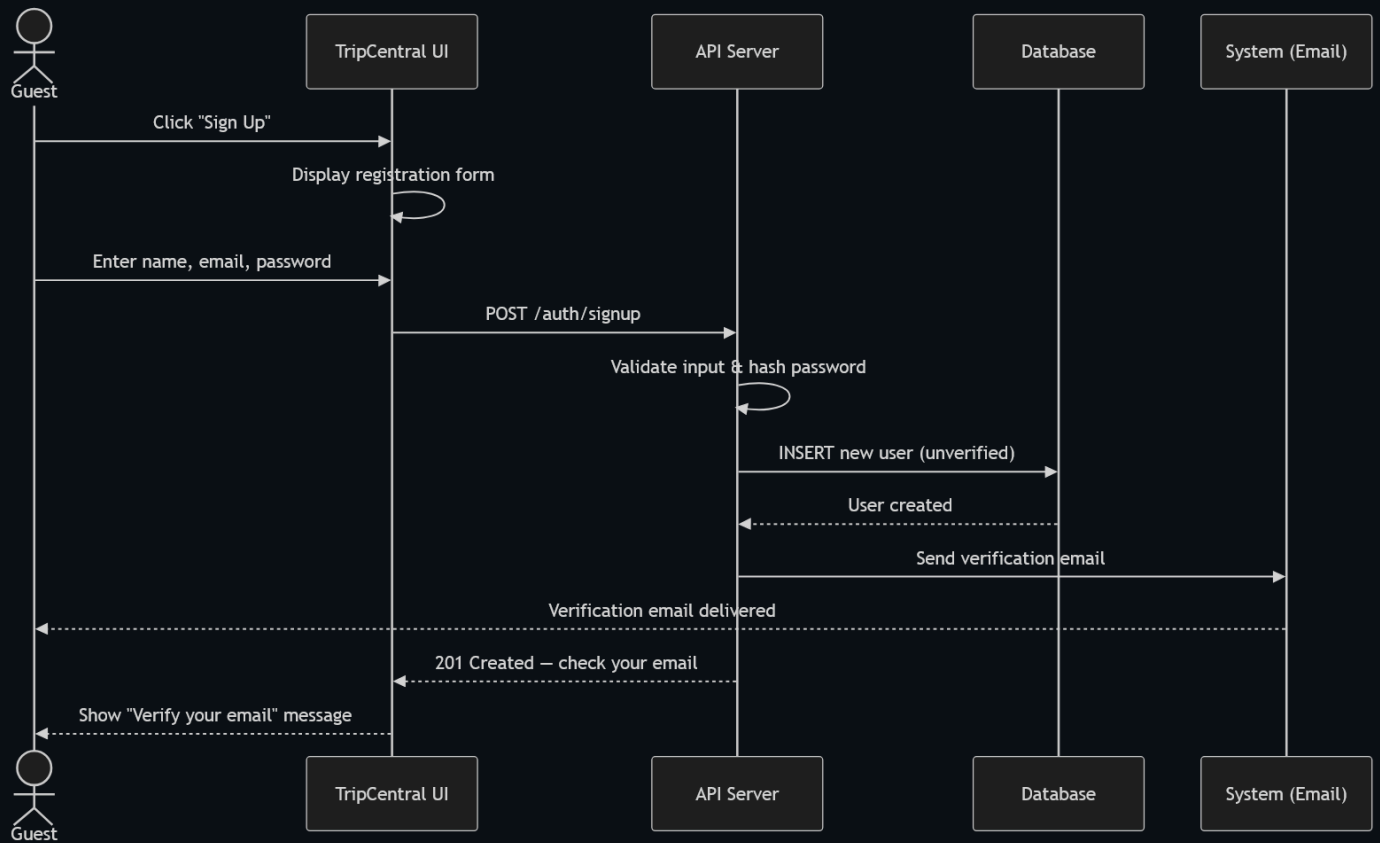
2.3 Use-case Diagram:



2.4 Sequence Diagram:

1. Sign Up

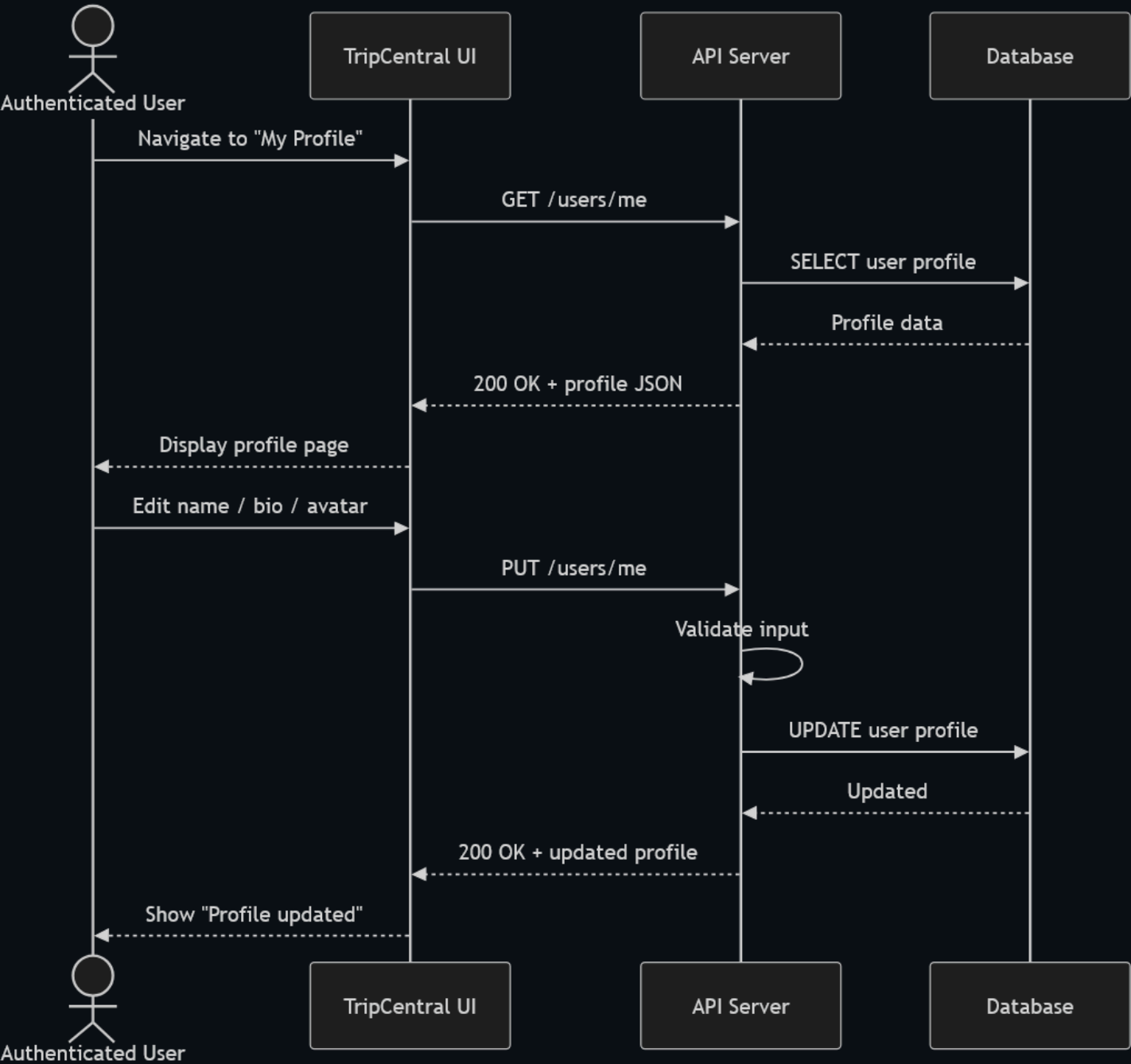
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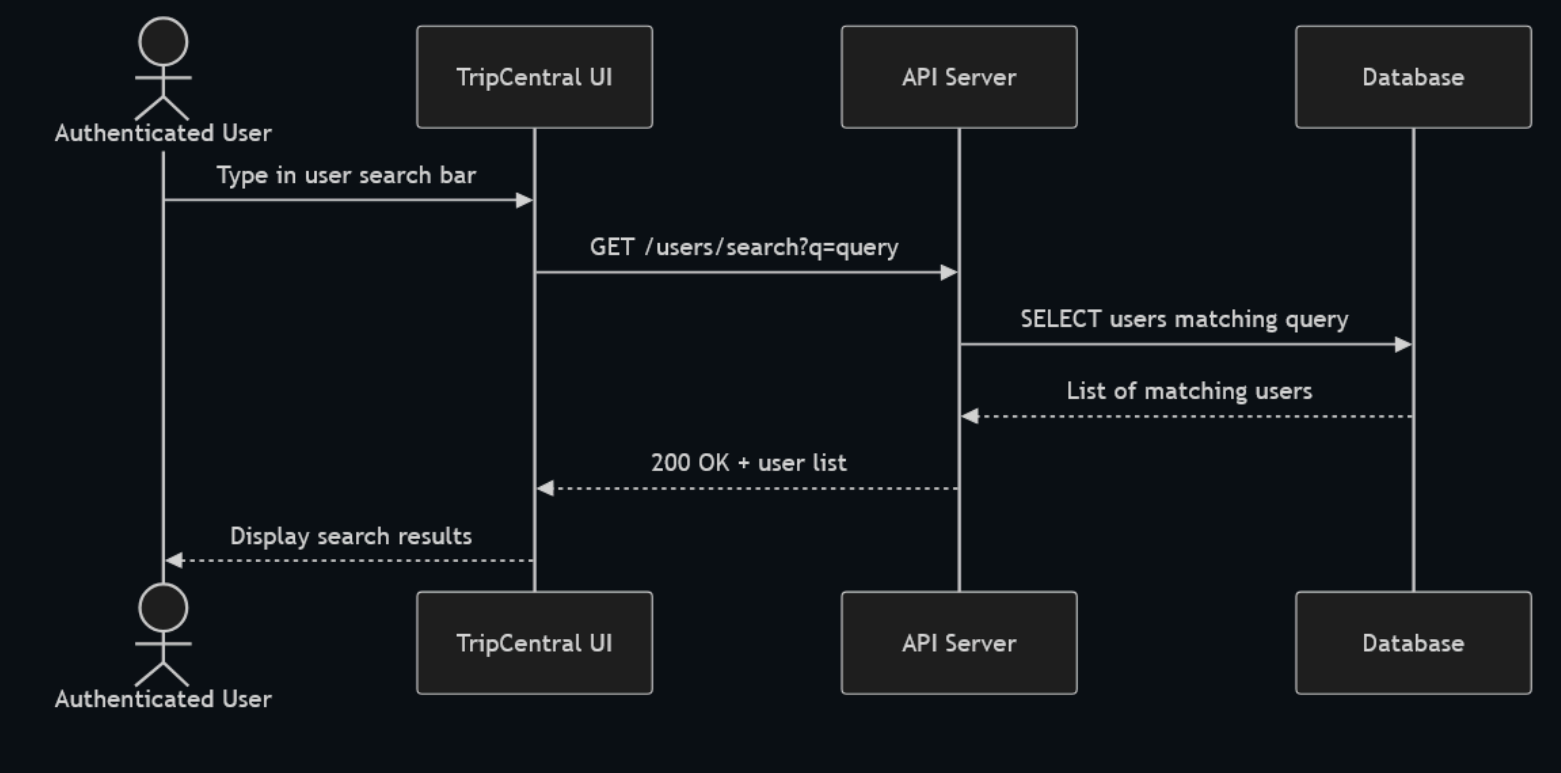
2. Log In



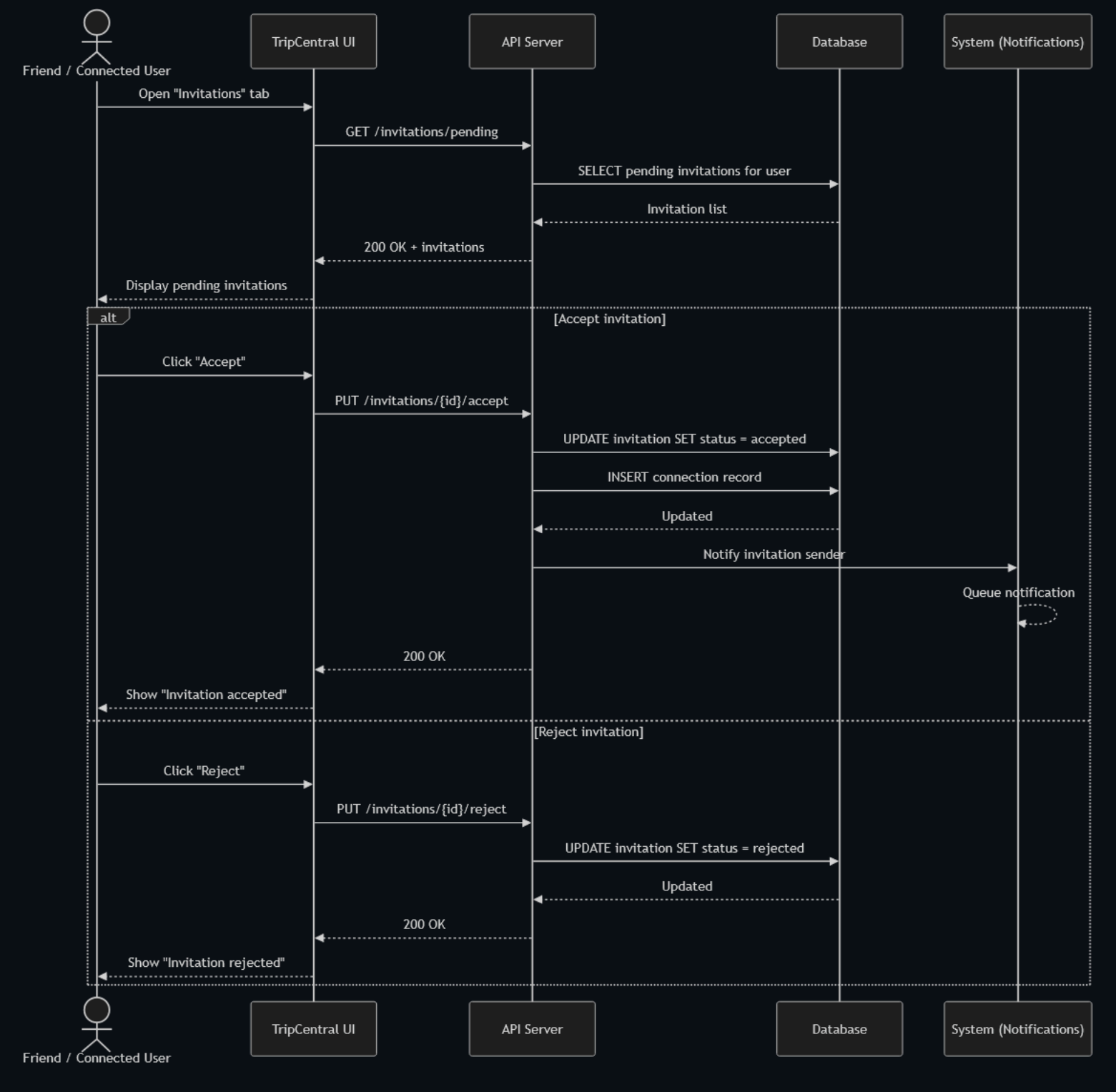
View / Edit Profile



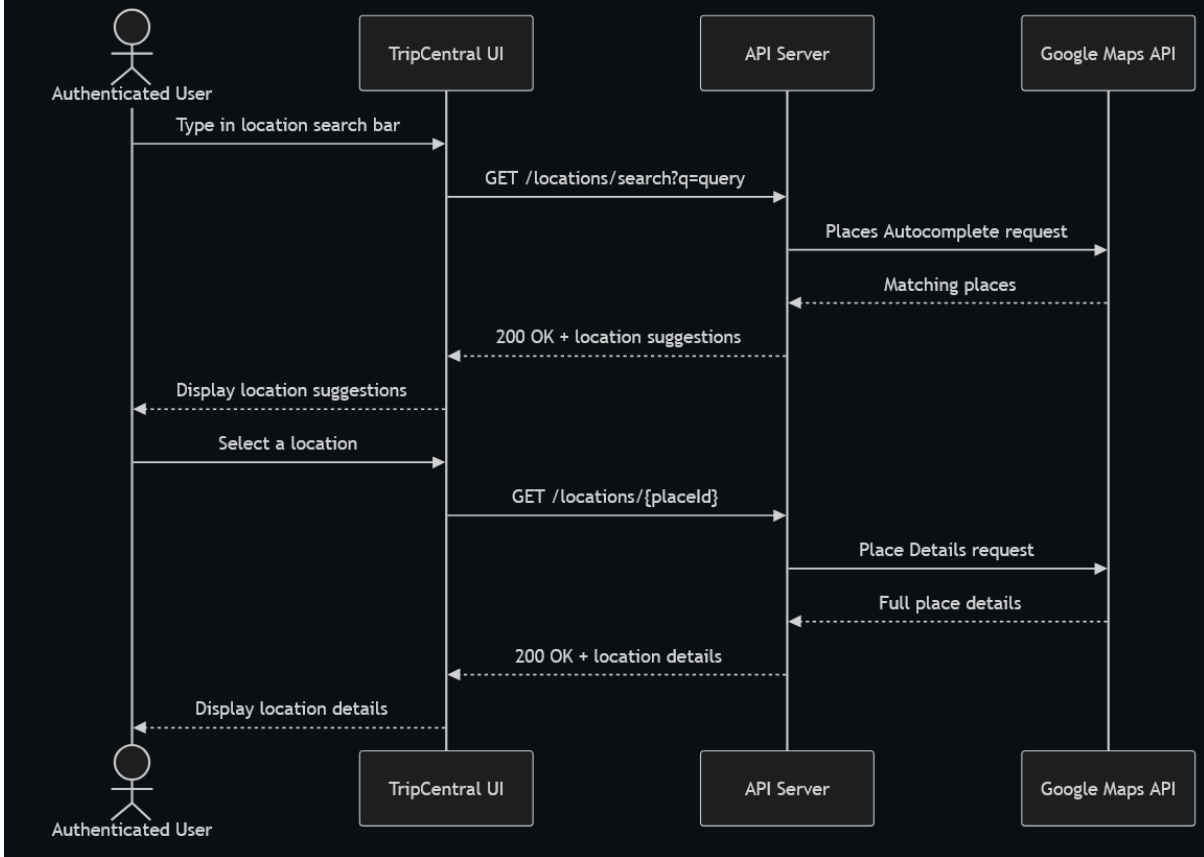
Search Users



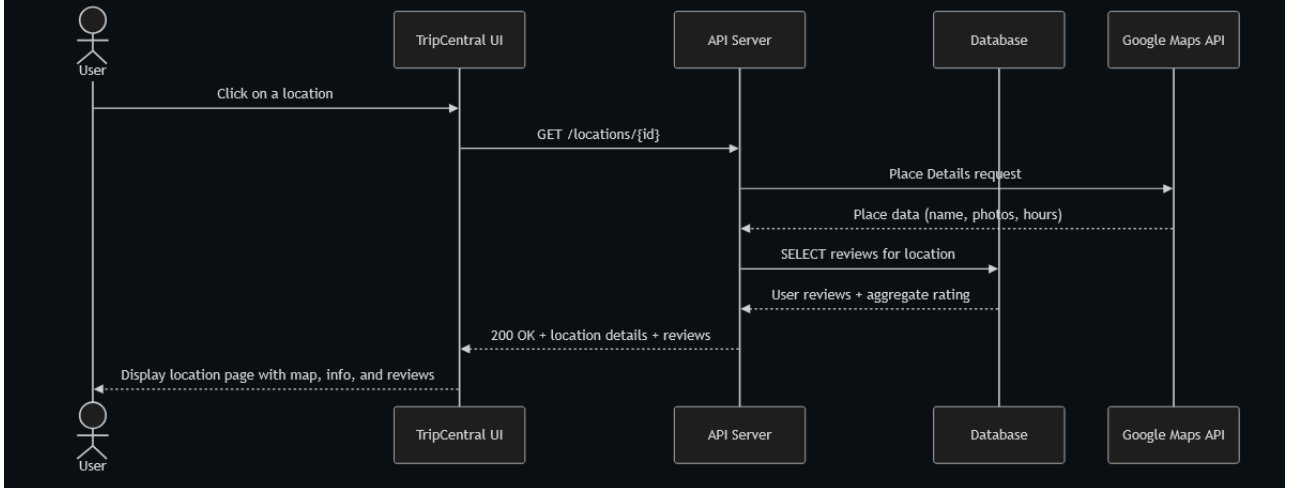
Accept / Reject Invitation



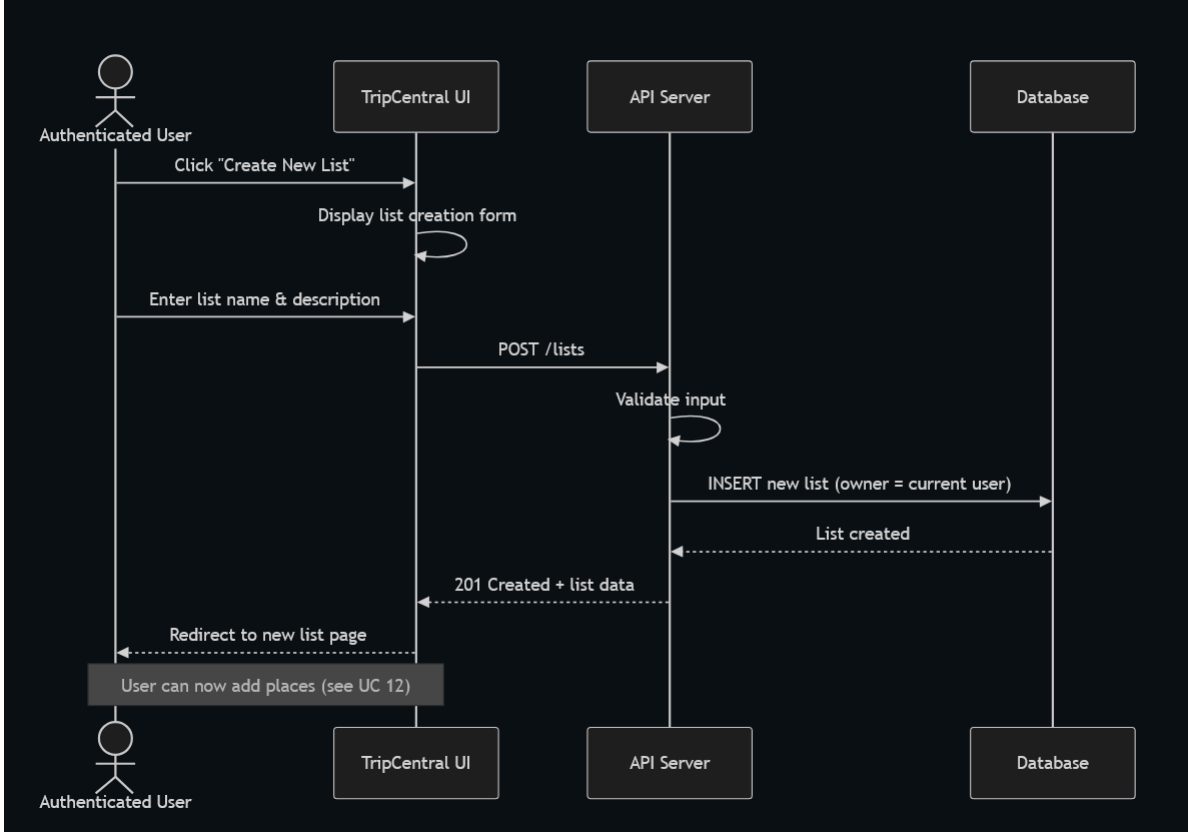
Search Locations (Google Maps)



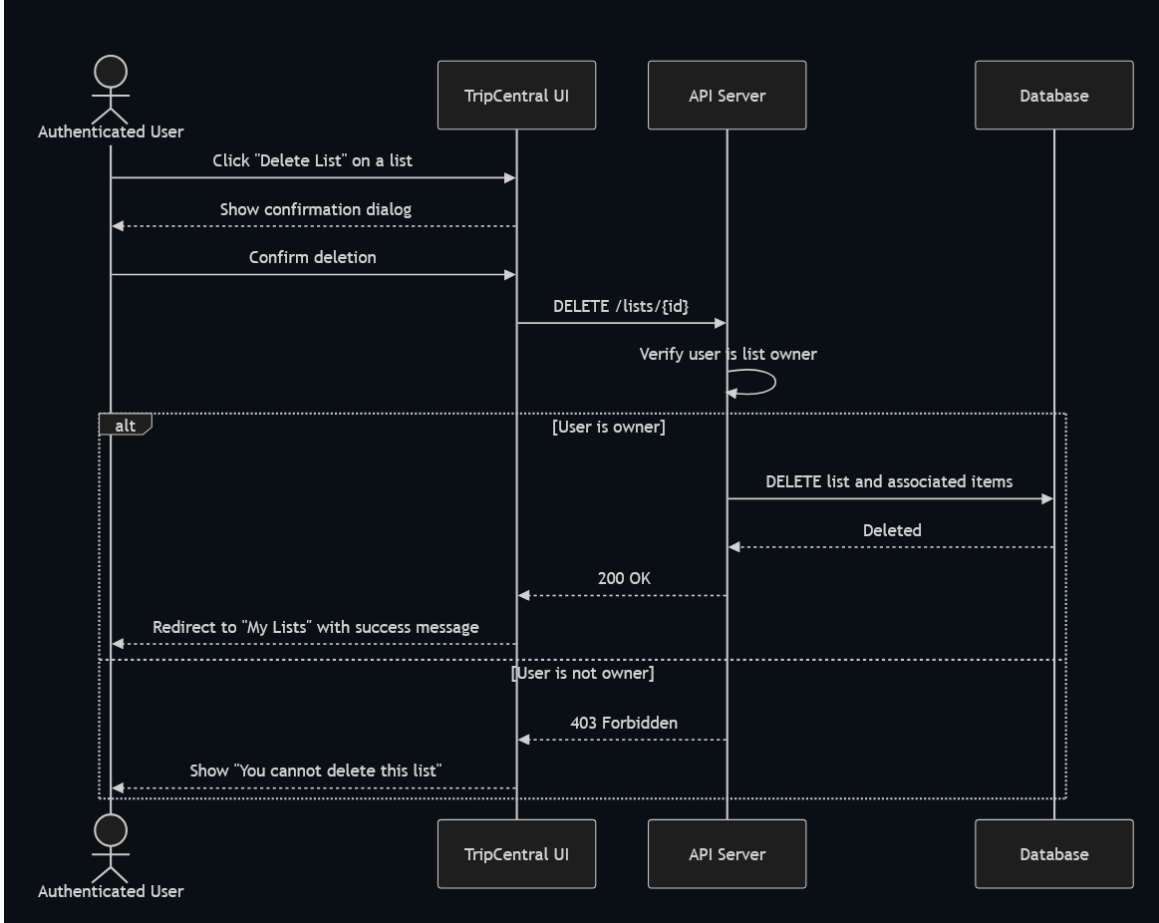
View Location Details



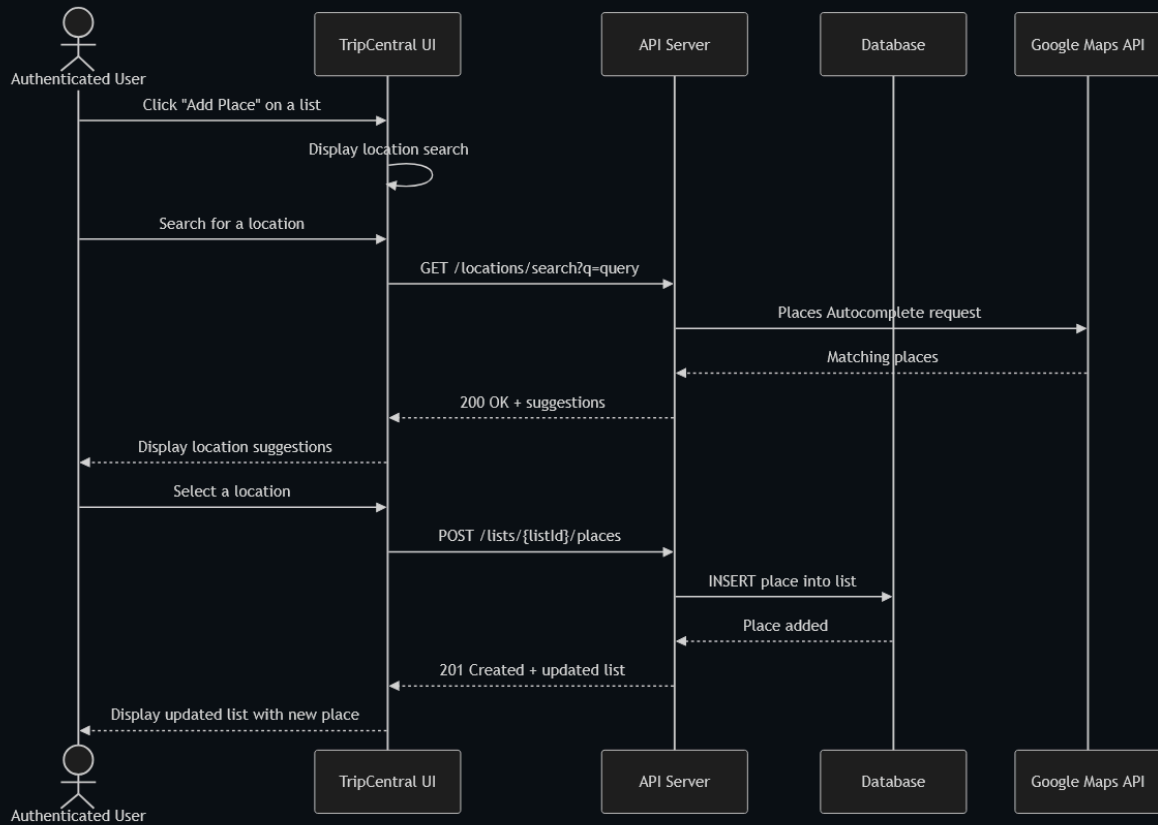
Create List



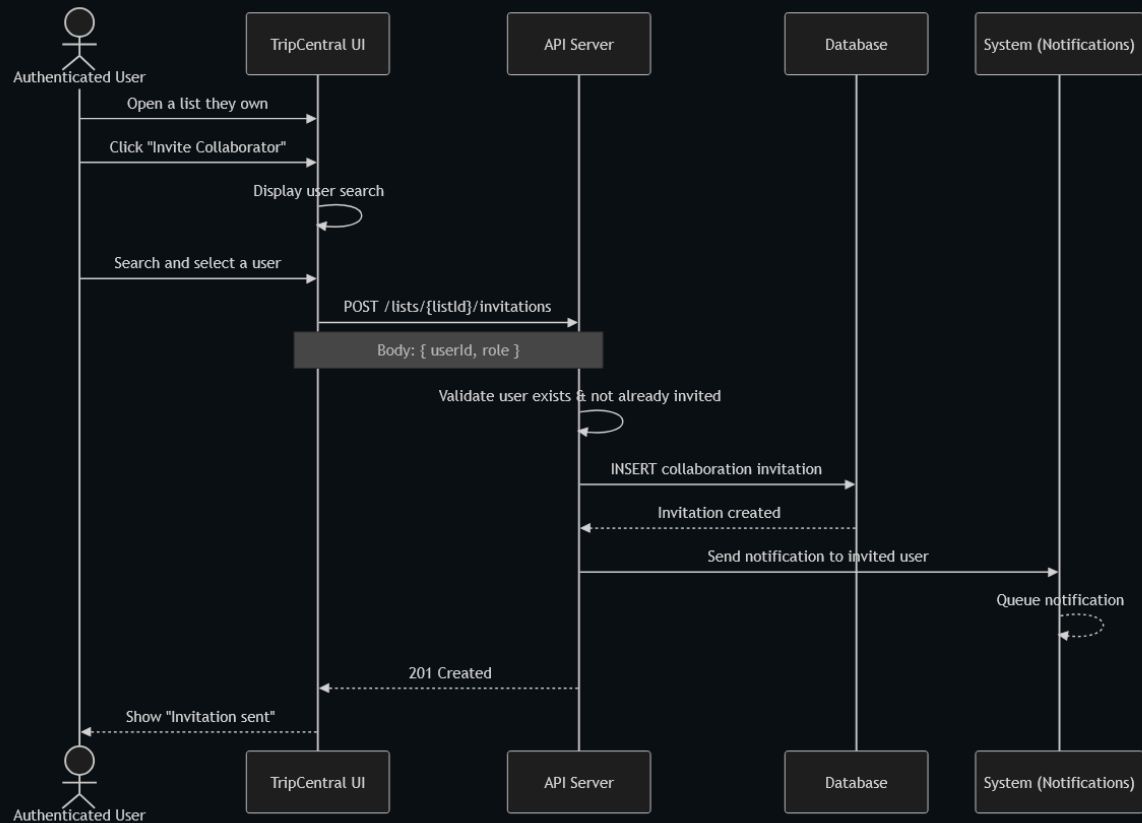
Delete List



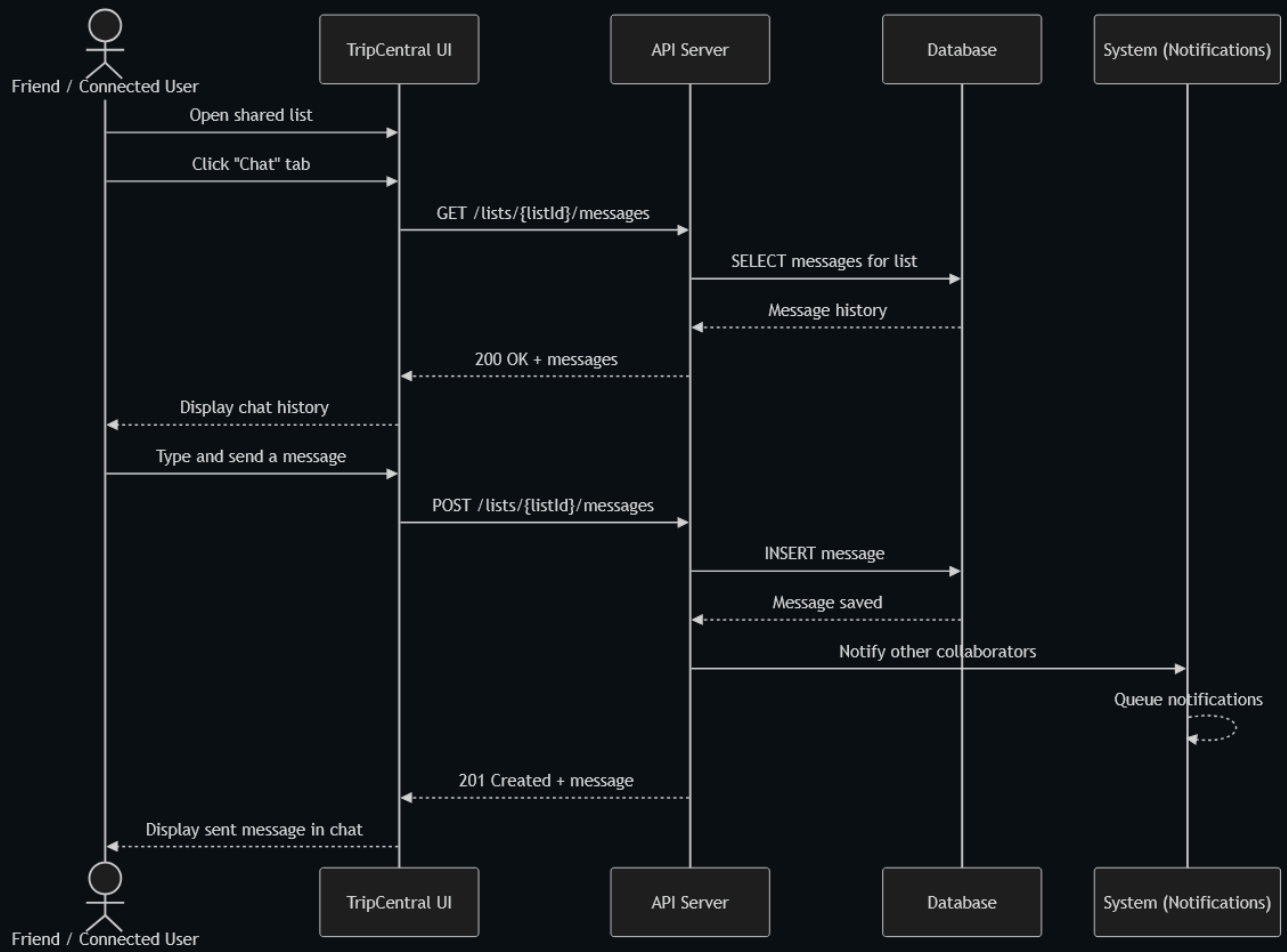
Add Place to List



Invite User to Collaborate



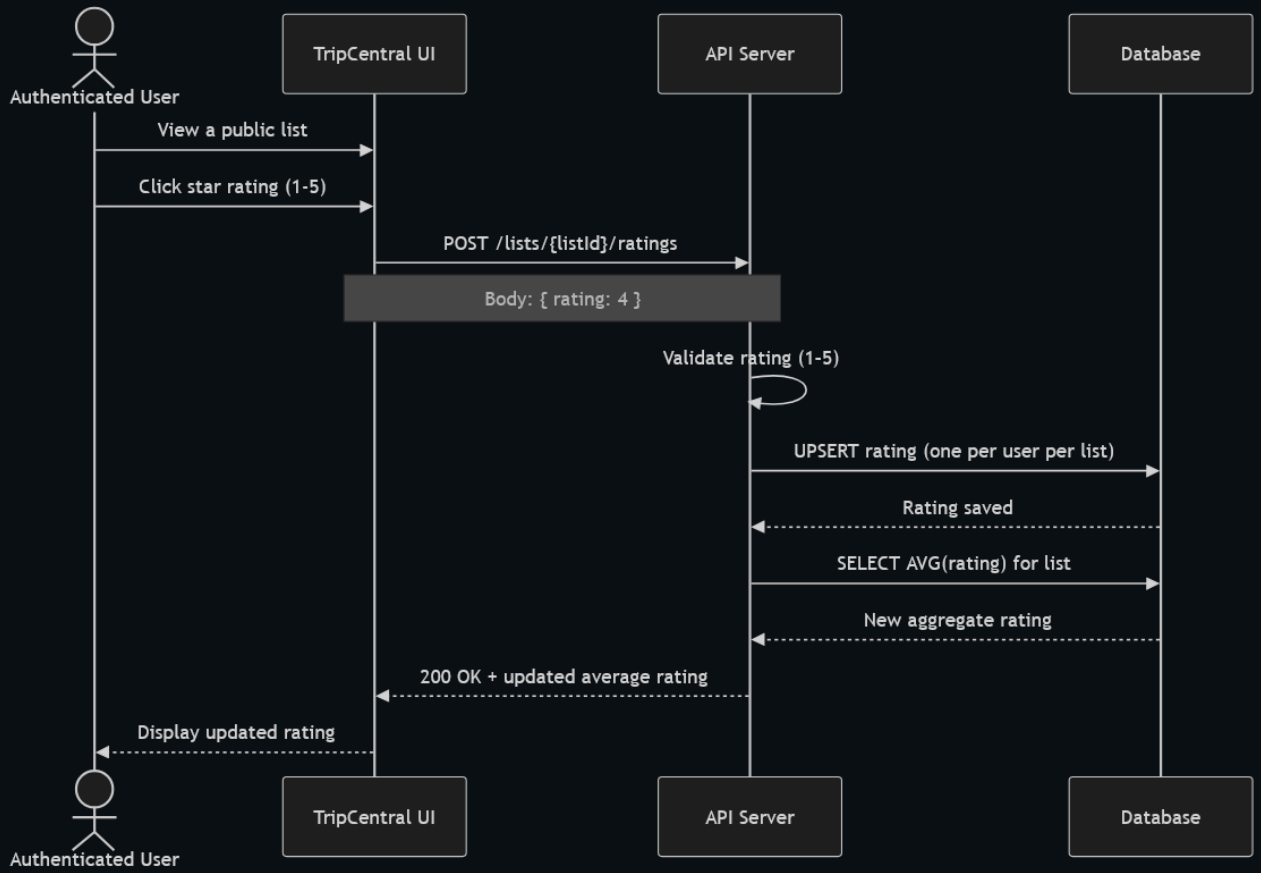
Chat with Collaborators



Browse Public Lists



Rate a List



Submit Place Review

